

Normalization

- 1.** What is normalization?
 - 2.** Why is normalization important?
 - 3.** What are the different normal forms?
 - 4.** Explain First Normal Form (1NF).
 - 5.** Explain Second Normal Form (2NF).
 - 6.** Explain Third Normal Form (3NF).
 - 7.** What is Boyce-Codd Normal Form (BCNF)?
 - 8.** Difference between 3NF and BCNF?
 - 9.** What is denormalization?
 - 10.** What are the advantages and disadvantages of normalization?
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◆ Functional Dependency

- 11.** What is functional dependency?
 - 12.** What is trivial functional dependency?
 - 13.** What is non-trivial functional dependency?
 - 14.** What is full functional dependency?
 - 15.** What is partial dependency?
 - 16.** What is transitive dependency?
 - 17.** How do you identify functional dependencies in a table?
 - 18.** What is the role of functional dependency in normalization?
 - 19.** Give an example of functional dependency.
 - 20.** What problems arise due to improper dependencies?
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◆ Types of Dependencies

- 21.** What is partial dependency?

- 22.** What is full functional dependency?
 - 23.** What is transitive dependency?
 - 24.** What is multi-valued dependency?
 - 25.** What is join dependency?
 - 26.** Difference between functional dependency and multi-valued dependency?
 - 27.** What is the impact of transitive dependency on normalization?
 - 28.** How do you remove partial dependency?
 - 29.** How do you remove transitive dependency?
 - 30.** What is Fourth Normal Form (4NF)?
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◆ Keys in DBMS

- 31.** What is a key in DBMS?
 - 32.** What is a primary key?
 - 33.** What is a candidate key?
 - 34.** What is a super key?
 - 35.** What is a composite key?
 - 36.** Difference between candidate key and super key?
 - 37.** Can a table have multiple candidate keys?
 - 38.** Can a primary key be composite?
 - 39.** What is a foreign key?
 - 40.** What is an alternate key?
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◆ Association (Relationships)

- 41.** What is association in DBMS?
- 42.** Types of relationships (1:1, 1:M, M:N)?
- 43.** What is cardinality?
- 44.** What is the participation constraint?

45. Difference between association and aggregation?

◆ **Scenario-Based Questions**

- 46.** Given a table, identify candidate keys.
- 47.** How do you normalize a table to 3NF?
- 48.** Identify and remove partial dependency from a table.
- 49.** Identify transitive dependency in a given schema.
- 50.** How do you convert a table from 2NF to 3NF?
- 51.** Design a table and normalize it step by step.
- 52.** Given functional dependencies, find the highest normal form.
- 53.** How do you handle multi-valued dependency?
- 54.** What happens if normalization is not applied?
- 55.** How do keys help in maintaining data integrity?

◆ **VIEW**

- 31. What is a view in SQL?
- 32. Why are views used?
- 33. Difference between view and table?
- 34. Can we update data through a view?
- 35. What is a materialized view?
- 36. How do you create a view?
- 37. How do you drop a view?
- 38. Can a view be based on another view?
- 39. What are the advantages of views?
- 40. What are the limitations of views?

◆ **INDEX**

- 41. What is an index in SQL?
- 42. Why is indexing used?
- 43. Types of indexes?
- 44. Difference between clustered and non-clustered index?
- 45. Can we create multiple indexes on a table?
- 46. What are disadvantages of indexing?
- 47. What happens if the index is not used?

48. How do you create an index?
49. How do you remove an index?
50. What is a composite index?

◆ Scenario-Based Questions

51. Which is faster: IN or EXISTS? When?
52. How would you optimize a slow query using indexes?
53. How do joins impact performance?
54. When would you avoid using indexes?
55. How do you handle duplicate records using joins?

◆ Subquery (Nested Query)

1. What is a subquery?
2. What are the types of subqueries?
3. What is the difference between correlated and non-correlated subquery?
4. Can a subquery return multiple columns?
5. Where can subqueries be used in SQL statements?
6. What is the difference between subquery and join?
7. What happens if a subquery returns more than one row with = operator?
8. Can we use subquery in FROM clause?
9. What is a scalar subquery?
10. Write a query using a subquery to find the second highest salary.

MCQ

1. Which normal form ensures atomic values in a table?

- A. 1NF
- B. 2NF
- C. 3NF
- D. BCNF

Answer: A

2. Which normal form removes partial dependency?

- A. 1NF
- B. 2NF
- C. 3NF
- D. BCNF

Answer: B

3. A table in 2NF but not in 3NF contains:

- A. Partial dependency
- B. Transitive dependency
- C. Multivalued dependency
- D. No dependency

Answer: B

4. BCNF is stronger than:

- A. 1NF
- B. 2NF
- C. 3NF
- D. 4NF

Answer: C

5. Functional dependency $X \rightarrow Y$ means:

- A. Y determines X
- B. X determines Y
- C. X and Y independent
- D. No relation

Answer: B

6. Which is a trivial functional dependency?

- A. $A \rightarrow B$
- B. $A \rightarrow A$
- C. $B \rightarrow C$
- D. $A \rightarrow C$

Answer: B

7. Partial dependency occurs when:

- A. Non-key depends on full key
- B. Non-key depends on part of composite key
- C. Key depends on non-key
- D. No dependency

Answer: B

8. Transitive dependency means:

- A. $A \rightarrow B$ and $B \rightarrow C$
- B. $A \rightarrow C$ directly
- C. $A \rightarrow B$ only
- D. $B \rightarrow A$

Answer: A

9. Which normal form removes transitive dependency?

- A. 2NF
- B. 3NF
- C. BCNF
- D. 4NF

Answer: B

10. Which normal form removes multivalued dependency?

- A. 2NF
- B. 3NF
- C. BCNF
- D. 4NF

Answer: D

11. What is a primary key?

- A. Duplicate key
- B. Unique identifier
- C. Foreign key
- D. Optional key

Answer: B

12. A table can have how many candidate keys?

- A. One
- B. Two
- C. Multiple
- D. None

Answer: C

13. Super key is:

- A. Minimal key
- B. Unique identifier set
- C. Foreign key
- D. Composite key

Answer: B

14. Composite key consists of:

- A. Single attribute
- B. Multiple attributes
- C. Only primary key
- D. Only foreign key

Answer: B

15. Foreign key is used to:

- A. Identify rows
- B. Link tables
- C. Delete data
- D. Sort data

Answer: B

16. Which SQL statement creates a view?

- A. CREATE TABLE
- B. CREATE VIEW
- C. MAKE VIEW
- D. ADD VIEW

Answer: B

17. A view is:

- A. Physical table
- B. Virtual table
- C. Index
- D. Key

Answer: B

18. Can a view be updated?

- A. Always
- B. Never
- C. Sometimes
- D. Only with index

Answer: C

19. Materialized view:

- A. Is virtual
- B. Stores data physically
- C. Cannot be queried
- D. Is temporary

Answer: B

20. Which command removes a view?

- A. DELETE VIEW
- B. REMOVE VIEW
- C. DROP VIEW
- D. ERASE VIEW

Answer: C

21. Index is used to:

- A. Slow down queries
- B. Improve performance
- C. Delete records
- D. Normalize table

Answer: B

22. Clustered index:

- A. Logical order
- B. Physical order
- C. No order
- D. Temporary

Answer: B

23. Can a table have multiple indexes?

- A. No
- B. Yes
- C. Only one
- D. Only primary key

Answer: B

24. Disadvantage of indexing:

- A. Faster SELECT
- B. Slower INSERT
- C. Less storage
- D. No effect

Answer: B

25. Create index syntax:

- A. CREATE INDEX idx ON table(col)
- B. MAKE INDEX
- C. ADD INDEX
- D. NEW INDEX

Answer: A

26. Subquery is:

- A. Nested query
- B. Table
- C. Index
- D. Constraint

Answer: A

27. Correlated subquery:

- A. Executes once
- B. Executes per row
- C. No relation
- D. Only join

Answer: B

28. If subquery returns multiple rows with '=':

- A. Works
- B. Error
- C. Ignores
- D. Converts

Answer: B

29. Scalar subquery returns:

- A. Table
- B. Multiple rows
- C. Single value
- D. Column

Answer: C

30. Query to find second highest salary uses:

- A. Join
- B. Subquery
- C. Index
- D. View

Answer: B

Aptitude MCQ

Q1.

A shopkeeper buys an item for ₹100 and sells it for ₹120. Find profit %.

Options:

A. 10% | B. 15% | C. 20% | D. 25%

Answer: C

Solution (Step by Step):

1. Profit = SP – CP = 120 – 100 = 20
 2. Profit % = (Profit ÷ CP) × 100 = (20 ÷ 100) × 100 = 20%
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Q2.

CP = ₹200, Loss = ₹50. Find Loss %.

Options:

A. 20% | B. 25% | C. 30% | D. 35%

Answer: B

Solution:

1. $\text{Loss \%} = (\text{Loss} \div \text{CP}) \times 100 = (50 \div 200) \times 100 = 25\%$

Q3.

If SP = ₹150 and profit = 50%, find CP.

Options:

A. ₹100 | B. ₹120 | C. ₹90 | D. ₹110

Answer: A

Solution:

1. Profit % = 50% $\rightarrow$ SP = 150% of CP
 2. $\text{CP} = \text{SP} \div 1.5 = 150 \div 1.5 = ₹100$
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Q4.

An item is sold at ₹80 with 20% loss. Find CP.

Options:

A. ₹90 | B. ₹95 | C. ₹100 | D. ₹110

Answer: C

Solution:

1. Loss = 20% $\rightarrow$ SP = 80% of CP
 2. $\text{CP} = \text{SP} \div 0.8 = 80 \div 0.8 = ₹100$
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Q5.

Profit = ₹40, CP = ₹160. Find SP.

Options:

A. ₹180 | B. ₹200 | C. ₹210 | D. ₹190

Answer: B

Solution:

1. $\text{SP} = \text{CP} + \text{Profit} = 160 + 40 = ₹200$
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Q6.

If profit % = 25% and CP = ₹400, find SP.

Options:

A. ₹450 | B. ₹500 | C. ₹480 | D. ₹520

Answer: B

Solution:

1. Profit = 25% of 400 = 100
 2. SP = CP + Profit = 400 + 100 = 500
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Q7.

A trader marks 20% above CP and gives 10% discount. Find profit %.

Options:

A. 8% | B. 10% | C. 12% | D. 15%

Answer: A

Solution:

1. Marked Price = 120% of CP
 2. After 10% discount $\rightarrow$ SP = 90% of 120 = 108% of CP
 3. Profit % = 108 - 100 = 8%
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Q8.

CP = ₹250, SP = ₹300. Find profit.

Options:

A. ₹40 | B. ₹50 | C. ₹60 | D. ₹70

Answer: B

Solution:

1. Profit = SP - CP = 300 - 250 = ₹50
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Q9.

Loss % = 10%, SP = ₹180. Find CP.

Options:

A. ₹190 | B. ₹200 | C. ₹210 | D. ₹220

Answer: B

Solution:

1. SP = 90% of CP $\rightarrow$ CP = SP $\div$ 0.9 = 180 $\div$ 0.9 = ₹200
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Q10.

Profit % = 50%, CP = ₹300. Find SP.

Options:

A. ₹400 | B. ₹450 | C. ₹500 | D. ₹550

Answer: B

Solution:

1. Profit = 50% of 300 = 150
2. SP = CP + Profit = 300 + 150 = 450

Q11.

SI on ₹1000 at 10% for 2 years.

Options:

A. ₹100 | B. ₹200 | C. ₹300 | D. ₹400

Answer: B

Solution:

1. $SI = (P \times R \times T)/100 = (1000 \times 10 \times 2)/100 = 200$
-

Q12.

Find total amount (SI) for above.

Options:

A. ₹1100 | B. ₹1200 | C. ₹1300 | D. ₹1400

Answer: B

Solution:

1. Amount = P + SI = 1000 + 200 = 1200
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Q13.

CI on ₹1000 at 10% for 2 years.

Options:

A. ₹200 | B. ₹210 | C. ₹220 | D. ₹230

Answer: B

Solution:

1. Year 1 $\rightarrow 1000 \times 1.1 = 1100$
 2. Year 2 $\rightarrow 1100 \times 1.1 = 1210$
 3. CI = 1210 - 1000 = 210
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Q14.

Difference between CI & SI (above case).

Options:

A. ₹5 | B. ₹10 | C. ₹15 | D. ₹20

Answer: B

Solution:

1. $CI - SI = 210 - 200 = 10$

Q15.

SI formula.

Options:

A. $P+R+T$ | B. $(P \times R \times T)/100$ | C. $P \times T$ | D. $R \times T$

Answer: B

Solution:

1. $SI = (P \times R \times T)/100$

Q16.

Rate = 5%, P = 2000, T = 3. Find SI.

Options:

A. ₹200 | B. ₹250 | C. ₹300 | D. ₹350

Answer: C

Solution:

1. $SI = (2000 \times 5 \times 3)/100 = 300$

Q17.

Amount after 1 year (CI) on ₹1000 at 10%.

Options:

A. ₹1100 | B. ₹1200 | C. ₹1050 | D. ₹1000

Answer: A

Solution:

1. $CI = 1000 \times 10/100 = 100$
 2. Amount = $1000 + 100 = 1100$
-

Q18.

If CI = SI, then time is

Options:

A. 1 year | B. 2 years | C. 3 years | D. 4 years

Answer: A

Solution:

1. CI = SI only in the first year (no compounding effect yet)
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Q19.

Principal = ₹500, SI = ₹50, Rate = 10%. Find time.

Options:

A. 1 year | B. 2 years | C. 3 years | D. 4 years

Answer: A

Solution:

1. $SI = (P \times R \times T)/100 \rightarrow 50 = (500 \times 10 \times T)/100$
 2. $50 = 50 \times T \rightarrow T = 1 \text{ year}$
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Q20.

CI grows faster than SI because

Options:

A. Higher rate | B. Compounding | C. Less time | D. Fixed rate

Answer: B

Solution:

1. CI adds interest on interest; SI only on principal
- Q21.

Speed formula.

Options:

A. $D \times T$ | B. $D \div T$ | C. $T \div D$ | D. $D + T$

Answer: B

Solution:

1. $\text{Speed} = \text{Distance} \div \text{Time}$
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Q22.

Distance = 100 km, Time = 2 hr. Speed?

Options:

A. 40 | B. 50 | C. 60 | D. 70

Answer: B

Solution:

1. $\text{Speed} = 100 \div 2 = 50 \text{ km/hr}$
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Q23.

Time = Distance $\div$ Speed?

Options:

A. True | B. False | C. Depends | D. None

Answer: A

Solution:

1. Time = Distance $\div$ Speed
-

Q24.

If speed doubles, time becomes?

Options:

A. Double | B. Half | C. Same | D. Zero

Answer: B

Solution:

1. Time $\propto$ 1/Speed $\rightarrow$ If speed doubles, time halves
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Q25.

60 km/hr = ? m/s

Options:

A. 16.67 | B. 20 | C. 25 | D. 15

Answer: A

Solution:

1. Multiply by 5/18 $\rightarrow 60 \times 5 \div 18 = 16.67$ m/s
-

Q26.

10 m/s = ? km/hr

Options:

A. 18 | B. 36 | C. 20 | D. 25

Answer: B

Solution:

1. Multiply by 18/5 $\rightarrow 10 \times 18 \div 5 = 36$ km/hr
-

Q27.

Distance covered in 3 hrs at 40 km/hr?

Options:

A. 100 | B. 120 | C. 140 | D. 150

Answer: B

Solution:

1. Distance = Speed $\times$ Time = $40 \times 3 = 120$ km

Q28.

A train travels 300 km in 5 hrs. Speed?

Options:

A. 50 | B. 60 | C. 70 | D. 80

Answer: B

Solution:

1. Speed = $300 \div 5 = 60$ km/hr

Q29.

If distance fixed, speed $\uparrow$ then time $\downarrow$

Options:

A. True | B. False | C. Depends | D. None

Answer: A

Solution:

1. Time $\propto 1/\text{Speed} \rightarrow$ Increasing speed reduces time

Q30.

Formula for distance

Options:

A. $S \div T$ | B. $S \times T$ | C. $T \div S$ | D. $S + T$

Answer: B

Solution:

1. Distance = Speed $\times$ Time

◆ VIEW (20 Problems)

1.

Create a view `emp_view` to display all columns from Employee table.

2.

Create a view `high_salary_emp` for employees with salary > 50000.

3.

Create a view `dept10_emp` for employees in department 10.

4.

Create a view `emp_name_salary` showing only name and salary.

5.

Create a view `emp_bonus` adding 10% bonus column.

6.

Create a view `emp_dept_view` using join between Employee and Department.

7.

Create a view `avg_salary_view` showing department-wise average salary.

8.

Create a view `top_earners` for employees with salary above average.

9.

Create a view `emp_count_dept` showing employee count per department.

10.

Create a view `emp_sorted` sorted by salary descending.

11.

Create a view recent_emp for employees hired after 2020.

12.

Create a view unique_dept showing distinct departments.

13.

Create a view emp_alias_view using column aliases.

14.

Create a view emp_view2 based on another view.

15.

Create a view emp_null_salary where salary is NULL.

16.

Create a view emp_range_salary for salary between 30000 and 70000.

17.

Create a view emp_group_view using GROUP BY department.

18.

Create a view emp_max_salary showing max salary per department.

19.

Create a view emp_min_salary showing minimum salary per department.

20.

Create a view emp_join_three joining Employee, Department, and Location tables.

◆ INDEX (20 Problems)

21.

Create an index `idx_emp_name` on `Employee(Name)`.

22.

Create an index `idx_salary` on `Employee(Salary)`.

23.

Create a composite index `idx_dept_salary` on `(DeptID, Salary)`.

24.

Create a unique index on `Email` column.

25.

Create an index on multiple columns `(Name, DeptID)`.

26.

Drop index `idx_emp_name`.

27.

Create an index on `Order` table for faster search by `OrderDate`.

28.

Create index on `Product` table for `ProductName`.

29.

Create index on foreign key column `DeptID`.

30.

Create index for optimizing join between `Employee` and `Department`.

31.

Create index on salary in descending order (if supported).

32.

Create index on Customer table for PhoneNumber.

33.

Drop index on Salary column.

34.

Create index on multiple columns (City, State).

35.

Create index on table with large dataset for performance testing.

36.

Create index on column frequently used in WHERE clause.

37.

Create index on column used in ORDER BY.

38.

Create index on column used in GROUP BY.

39.

Create index on nullable column.

40.

Create index on table to optimize LIKE 'A%' queries.

◆ SUBQUERY (20 Problems)

41.

Find employees whose salary is greater than average salary.

42.

Find employees with maximum salary using subquery.

43.

Find employees working in the same department as 'John'.

44.

Find employees not in department 10 using subquery.

45.

Find second highest salary using subquery.

46.

Find employees whose salary is less than minimum salary in another department.

47.

Find departments with more employees than average.

48.

Find employees whose salary equals max salary in their department.

49.

Use subquery in SELECT to display average salary.

50.

Use subquery in FROM clause to create derived table.

51.

Use correlated subquery to find employees earning above department average.

52.

Find employees who do not exist in another table.

53.

Use EXISTS to find employees in departments.

54.

Use NOT EXISTS to find employees without projects.

55.

Find employees whose salary is in top 3 salaries.

56.

Find employees with salary greater than all employees in department 10.

57.

Use subquery to delete employees with lowest salary.

58.

Use subquery in UPDATE to increase salary based on average.

59.

Use subquery with IN operator to filter records.

60.

Use scalar subquery to display department name along with employee.